

## Unit 2.7 The derivative as a

Note Title

2012/03/23

function.

Section 2.8 p 154 - 165

Def of the derivative function

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Example. Use the def of the derivative to find the gradient of the tangent line to the graph of  $f(x) = \sqrt{x+2}$  at  $x=1$ , and find the tangent line

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{x+h+2} - \sqrt{x+2}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{\sqrt{x+h+2} - \sqrt{x+2}}{h} \times \frac{\sqrt{x+h+2} + \sqrt{x+2}}{\sqrt{x+h+2} + \sqrt{x+2}}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h+2) - (x+2)}{h(\sqrt{x+h+2} + \sqrt{x+2})}$$

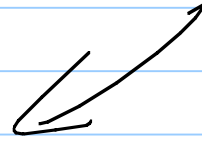
$$= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h+2} + \sqrt{x+2}} = \frac{1}{2\sqrt{x+2}} = f'(x)$$

$$(y - f(1)) = f'(1)(x - 1)$$

$$\Rightarrow (y - \sqrt{3}) = \frac{1}{2\sqrt{3}}(x - 1)$$

Example  $f(x) = \sqrt{x}$

Domain =  $[0, \infty)$

$f'(x) = \frac{1}{2} \frac{1}{\sqrt{x}}$   Use definition!

Domain =  $(0, \infty)$

- The domain of  $f' \subset \text{domain } f$

Example Find The derivative of The function using def

$$f(x) = \frac{x}{x+1} \quad \text{and find equation of}$$

tangent line at The point 1

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{x+h}{x+h+1} - \frac{x}{x+1}}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x+h)(x+1) - x(x+h+1)}{(x+h+1)(x+1)} \cdot h$$

$$= \lim_{h \rightarrow 0} \frac{1}{h} \left( \frac{x^2 + h + hx + x - (x^2 + xh + x)}{(x+h+1)(x+1)} \right)$$

$$= \lim_{h \rightarrow 0} \frac{1}{\cancel{h}} \left( \frac{\cancel{x^2} + \cancel{h} + \cancel{hx} + \cancel{x} - \cancel{x^2} - \cancel{xh} - \cancel{x}}{(x+h+1)(x+1)} \right)$$

$$= \lim_{h \rightarrow 0} \frac{1}{(x+h+1)(x+1)}$$

$$= \frac{1}{(x+1)^2} = f'(x)$$

$$(y - f(1)) = f'(1)(x - 1)$$

$$\left(y - \frac{1}{2}\right) = \frac{1}{4}(x - 1)$$

simplify



Homework Find the derivative function  
(using def) if

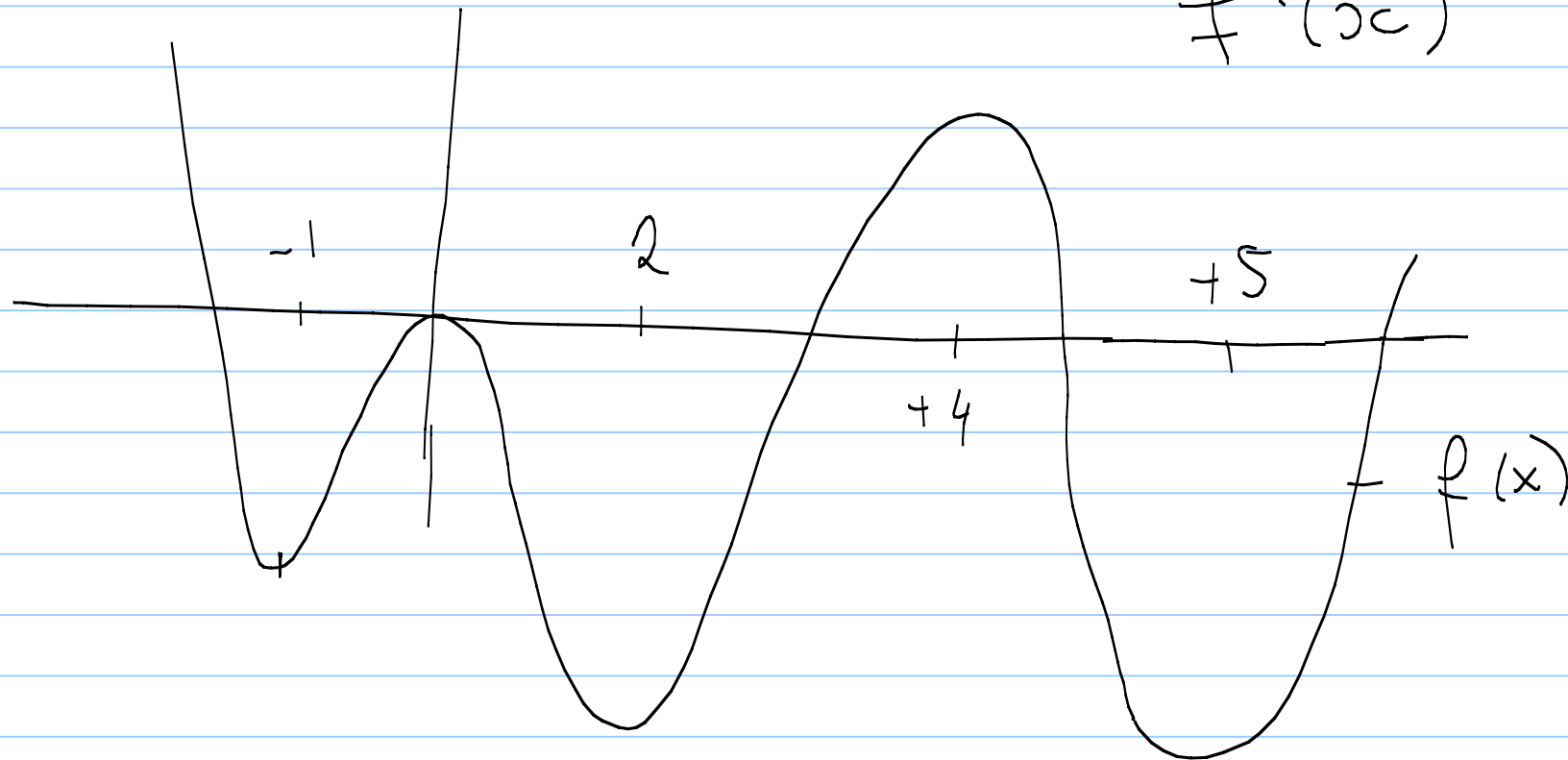
$$f(x) = x^3 - x.$$

The geometric meaning

$f$  is differentiable at  $x=a$   
if you can draw a tangent  
line with a finite slope at the  
point  $(a, f(a))$

Example

Given  $f(x)$ , sketch  
 $f'(x)$





Note:

\* Horizontal tangent line

Then  $f'(x) = 0$

\*  $f(x)$  increasing

Then  $f'(x) > 0$

\*  $f(x)$  decreasing  
Then  $f'(x) < 0$

Self study: differentiable on an interval.

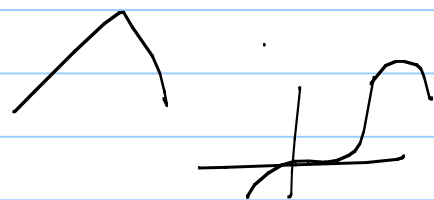
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Def A function  $f$  is differentiable at  $a$  if  $f'(x)$  exists. It is differentiable on an open interval  $(a, b)$  or  $(a, \infty)$  or  $(-\infty, a)$  or  $(-\infty, \infty)$  if it is differentiable at every number in the interval.

How can a function fail to be differentiable at a point?

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- ① If it has a corner
- ② a vertical tangent
- ③ it can be discontinuous at the point



Example Where is  $f(x) = |x|$   
differentiable?

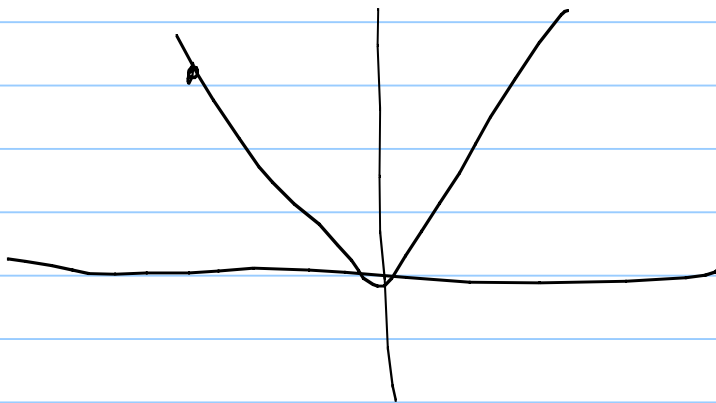
$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$



$$x < 0$$

$$\lim_{h \rightarrow 0^-} \frac{-(x+h) - (-x)}{h} = \lim_{h \rightarrow 0} \frac{-h}{h} = -1$$



If  $x > 0$

$$\lim_{h \rightarrow 0^+} \frac{(x+h) - (x)}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1.$$

For  $x=0$

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{|0+h| - |0|}{h}$$

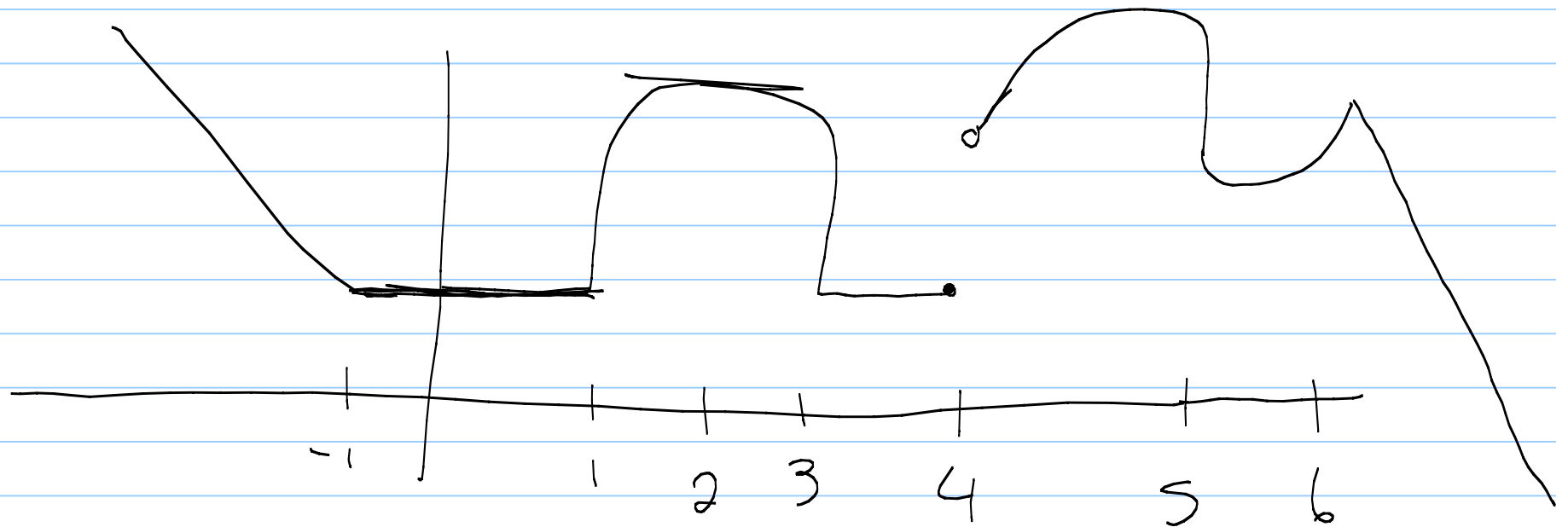
$$\lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$$

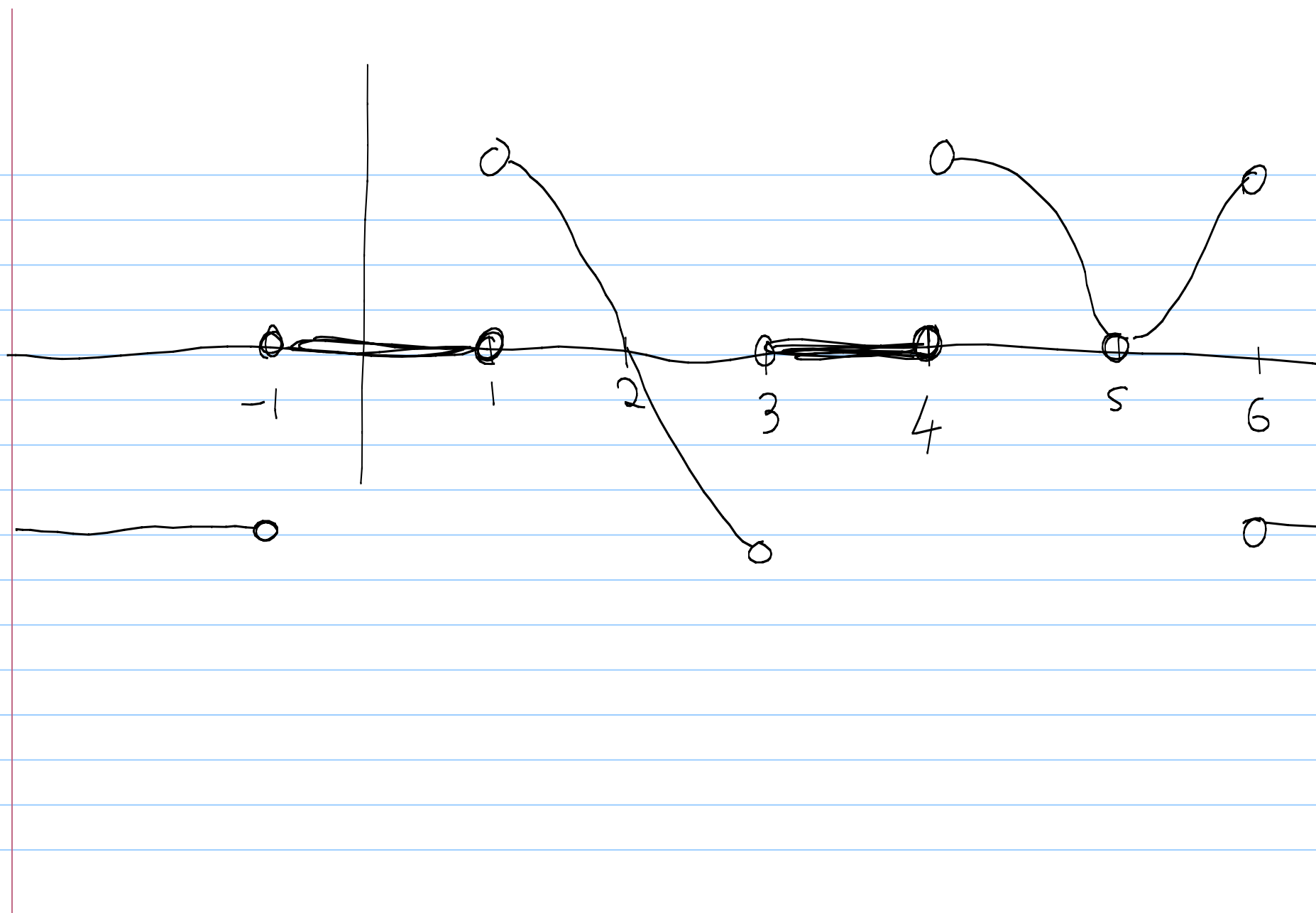
$$\lim_{h \rightarrow 0^+} \frac{h}{h} = 1$$

That  $\lim_{h \rightarrow 0^-} f(x) \neq \lim_{h \rightarrow 0^+} f(x)$

$\therefore f'(0)$  does not exist

Example. Given  $f(x)$ , sketch  $f'(x)$





Theorem 4 pg 15 (You need to  
know how to  
prove it!)

If  $f$  is differentiable at  $a$ ,  
then  $f$  is continuous at  $a$ .

Proof:  $f$  is differentiable.

$$\lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \text{ exists}$$

given

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \text{ exists}$$

prove:  $\lim_{x \rightarrow a} f(x) = f(a)$



$f(x) - f(a)$  approaches 0

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a} \cdot (x - a)$$

$$x \neq a.$$

Using the product law & The def  
of differentiability

$$\lim_{x \rightarrow a} [f(x) - f(a)] = \lim_{x \rightarrow a} \left[ \frac{f(x) - f(a)}{x - a} \cdot (x - a) \right]$$

$$= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \cdot \lim_{x \rightarrow a} (x - a)$$

$$= f'(a) \cdot (a - a) = f'(a) \cdot 0 = 0.$$

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} [f(x) + f(a) - f(a)]$$

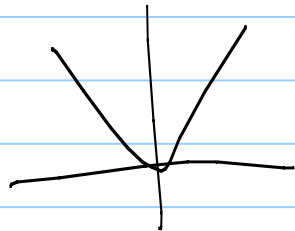
$$= \lim_{x \rightarrow a} [f(x) - f(a)] + \lim_{x \rightarrow a} f(a)$$

$$= 0 + f(a) = f(a)$$

NOTE: The converse of Theorem 4 is false!.

There are functions that are continuous but not differentiable.

Example:  $f(x) = |x|$



is continuous at  $x=0$   
but not differentiable

Example

$$f(x) = \begin{cases} x^2 & \text{if } x < 2 \\ -x + 6 & \text{if } x \geq 2 \end{cases}$$

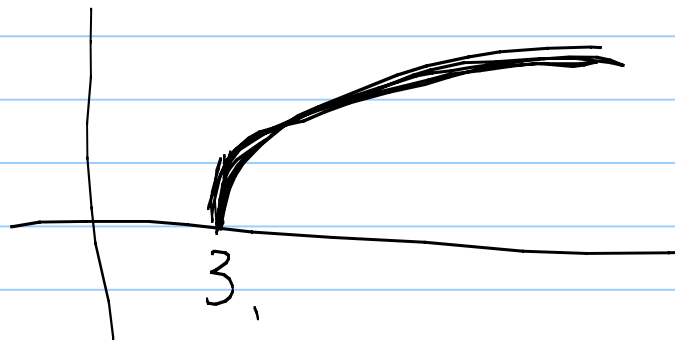
Is the function differentiable  
at  $x = 2$ ?

Use the def  $\lim_{x \rightarrow 2} \frac{f(x) - f(2)}{x - 2}$   
not diff.

Example (vertical tangent)

Is the function

$f(x) = (x-3)^{\frac{1}{3}}$  differentiable  
at  $x=3$ ?



Show

$\lim_{x \rightarrow 3} \frac{f(x) - f(3)}{x - 3}$   
does not exist.

Other notation for the derivative

If  $y = f(x)$  Then

$$\underline{f'(x) = D f(x) = D_x f(x)}$$

$$f'(x) = \frac{d}{dx} f(x) = \frac{df}{dx}$$

$$f'(x) = \frac{d}{dx} y = \frac{dy}{dx}$$

## Higher derivatives

$$f''(x) = \frac{d}{dx} f'(x)$$

$$= \frac{d^2}{dx^2} f(x)$$

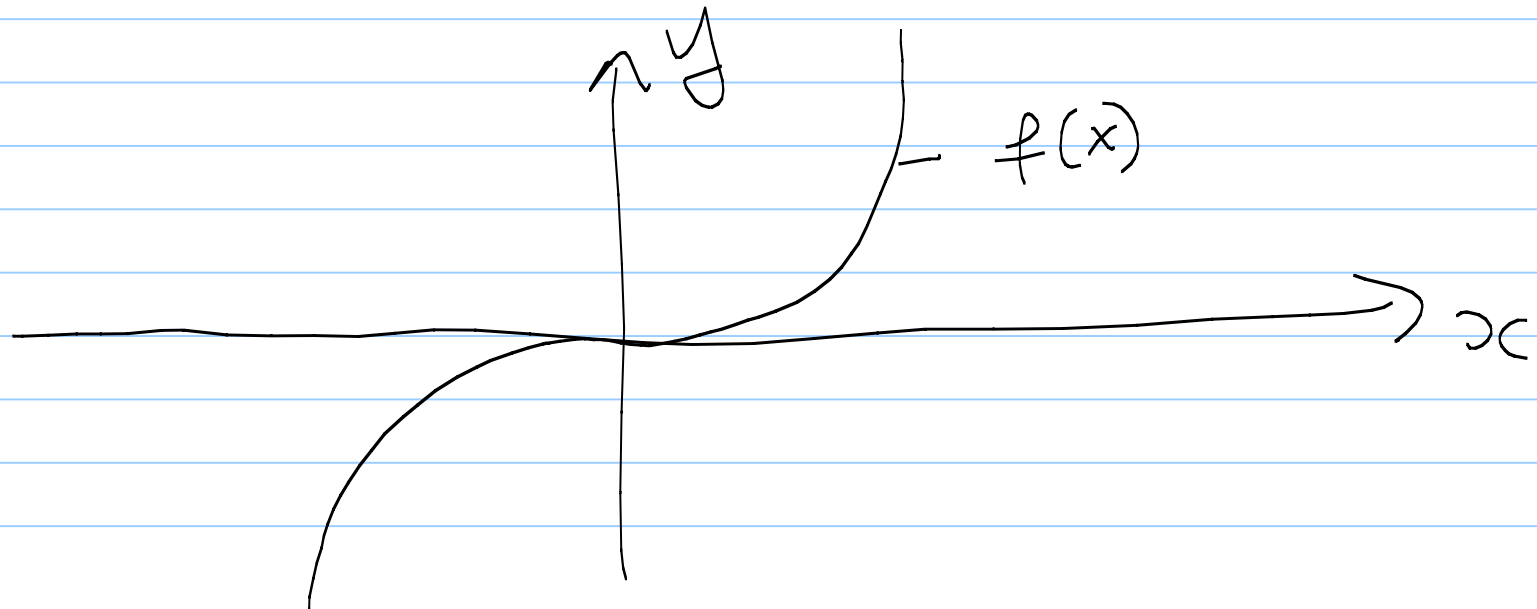
$$f^{(n)}(x) = \frac{d^n}{dx^n} f(x)$$

Example.

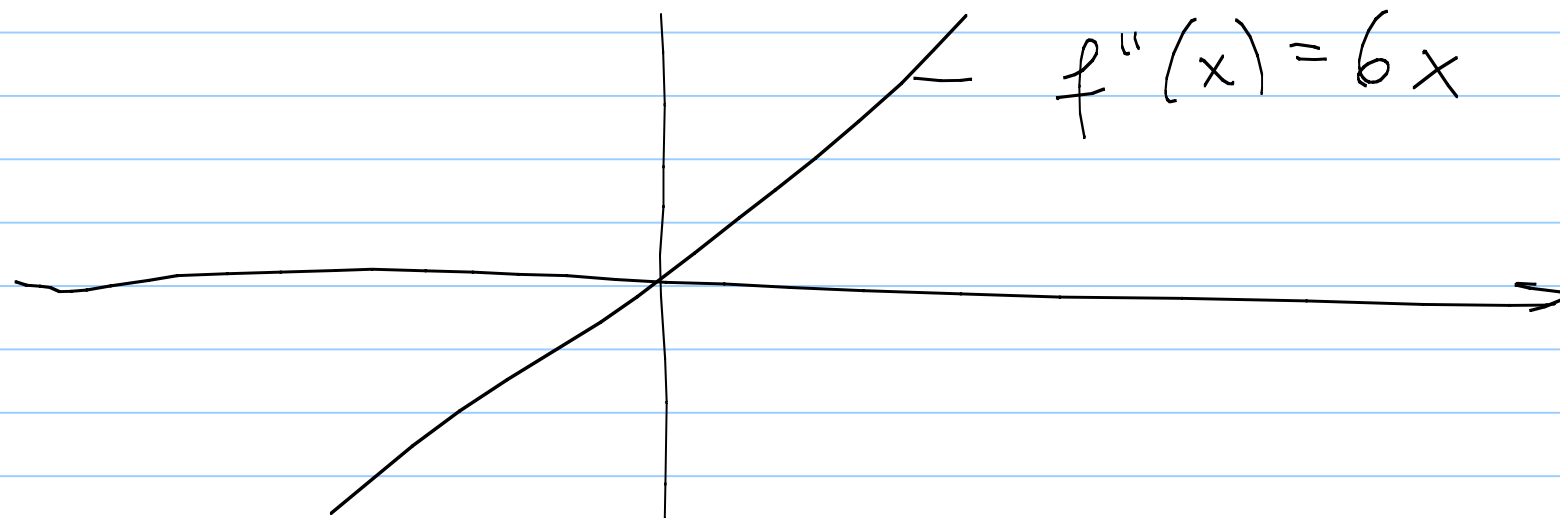
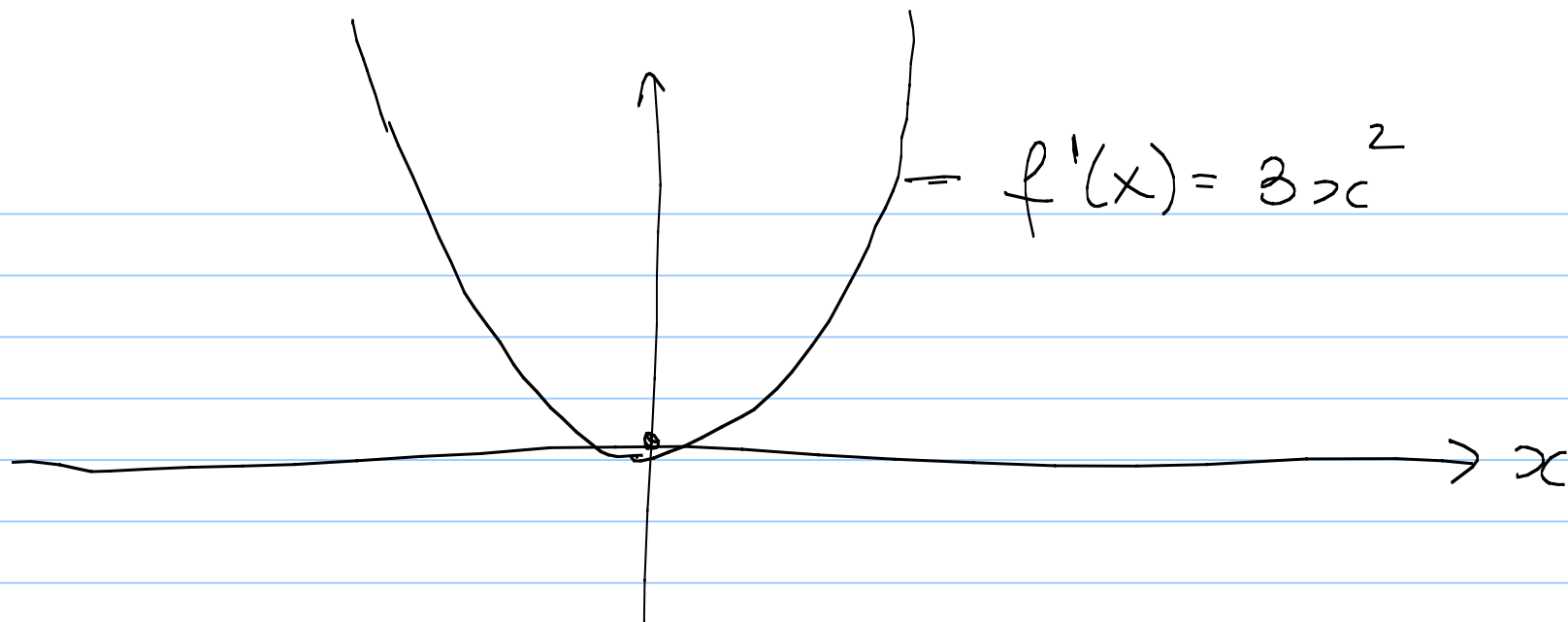
Sketch

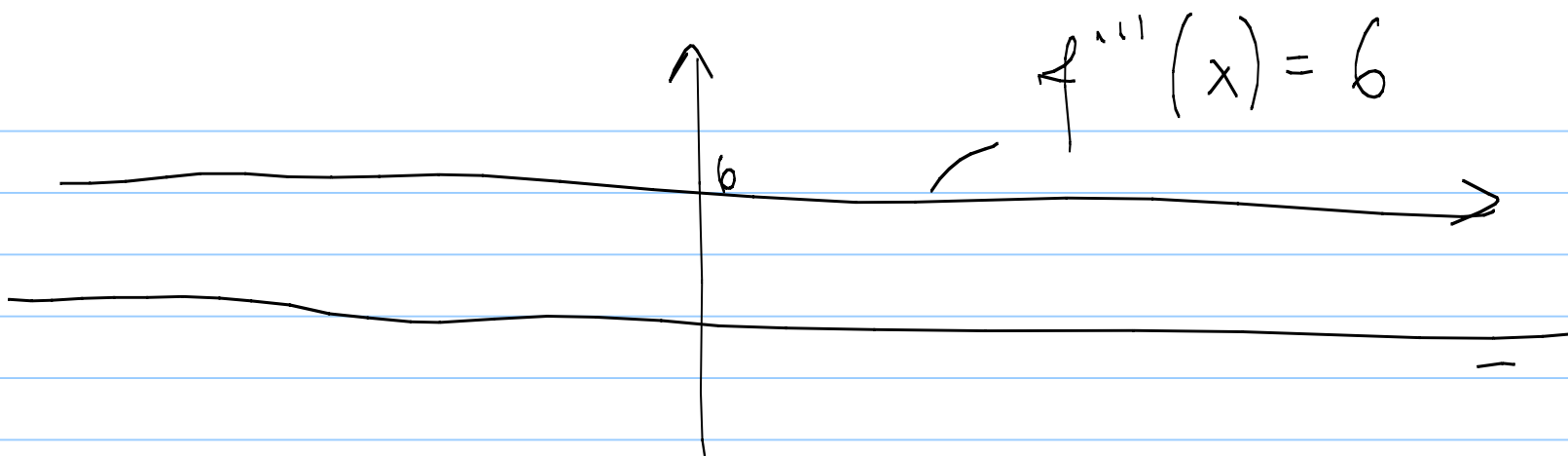
$f(x) = x^3$  and sketch

$f'(x)$  and  $f''(x)$  &  $f'''(x)$









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